Transformer equivalent circuit

Answers and worked solutions · Day 04 · Tue 27 Oct · pack M4

5. Practice

From your course files

Jun24 Q2b (past exam, 5 marks).

> This is worked in full as **Example 1** in section 3. Redo it without looking, then check.
>
> *Hint:* $Z_2 = n^2 Z_1$ and Z_3 adds the 15 Ω.
>
> i. $Z_1 =$ **0.3 Ω**, $Z_2 = 36 \times 0.3 =$ **10.8 Ω**, $Z_3 = 15 + 10.8 =$ **25.8 Ω**.
>
> ii. $I_1 = 12/25.8 =$ **0.4651 A**, $I_2 = 6I_1 =$ **2.791 A**, $V_1 = I_1 Z_2 =$ **5.023 V**, $V_2 = V_1/6 =$ **0.8372 V**.
>
> These match the official answers.

PQ 3.20 (practice questions).

> $n = 5$, $n^2 = 25$.
>
> a. $R'_L = n^2 R_L = 25 \times 10 =$ **250 Ω**.
>
> b. $I_1 = 240/(12 + 250) = 240/262 =$ **0.9160 A**.
>
> c. $V_1 = I_1 R'_L = 0.9160 \times 250 =$ **229.0 V**.
>
> d. $I_2 = nI_1 =$ **4.580 A** and $V_2 = V_1/n =$ **45.80 V**.
>
> e. $P_{OUT} = I_2^2 R_L = 4.580^2 \times 10 =$ **209.8 W**.
>
> f. $P_{12\Omega} = I_1^2 \times 12 =$ **10.07 W**.
>
> g. $\eta = 209.8/(240 \times 0.9160) = 209.8/219.8 =$ **95.42 %**.
>
> These match the official answers.

PQ 3.21 (practice questions).

> *Hint:* $5^2 \times 5 = 125\,\Omega$, then a divider with 12 Ω.
>
> $R'_L = 25 \times 5 = 125\ \Omega$. Divider: $V_1 = 240 \times \dfrac{125}{125 + 12} = 219.0$ V. Then $V_2 = V_1/5 =$ **43.8 V**.

> This matches the official answer.

Jun23 Q2b (past exam, 5 marks).

> *Hint:* SC $Q = 256.1$ VAr, OC $Q = 308.7$ VAr.
>
> **SC test (series R + jX).** $R = P/I^2 = 55/9.7^2 =$ **0.5845 Ω**. $|Z| = 27/9.7 = 2.784\ \Omega$, so $X = \sqrt{2.784^2 - 0.5845^2} =$ **2.721 Ω**. (Equivalently $Q = \sqrt{(27 \times 9.7)^2 - 55^2} = 256.1$ VAr and $X = Q/I^2$.)
>
> **OC test (parallel R_C and jX_M).** $R_C = V^2/P = 240^2/45 =$ **1280 Ω**.
> $S = 240 \times 1.3 = 312$ VA, so $Q = \sqrt{312^2 - 45^2} = 308.7$ VAr and
> $X_M = V^2/Q = 240^2/308.7 =$ **186.6 Ω**.
>
> All referred to the primary (the tests were measured there). These match the official answers.

Jun23 Q2c (past exam, 6 marks).

> i. $V_2' = V_1 - I_2'(R_1 + R_2' + jX_1 + jX_2')$.
>
> ii. $V_2' = V_1 \dfrac{R_L'}{R_L' + R_1 + R_2' + jX_1 + jX_2'}$. The shunt branch ($R_C$, jX_M) sits directly across V_1, so it doesn't enter the divider.
>
> iii. **Lagging.** The series reactance makes I_2' lag V_1, and V_2' is in phase with I_2' because the load is resistive.
>
> *Note:* the official sheet writes "V_1'" and "R_L". The primary voltage needs no prime, and the load should be the referred R_L' (the question defines it that way). This is a notation slip, not a different answer.

PQ 3.27 (practice questions).

> a. $n = 240/48 = 5$, $n^2 = 25$. $R = R_1 + n^2 R_2 = 0.6 + 25 \times 0.02 =$ **1.1 Ω**.
> $X = X_1 + n^2 X_2 = 2 + 25 \times 0.06 =$ **3.5 Ω**. The shunt branch is $R_C =$ **3200 Ω** in parallel with $X_M =$ **550 Ω**, placed across V_1.
>
> b. No load: only the shunt branch draws current. $I_R = 240/3200 = 0.075$ A and $I_M = 240/550 = 0.4364$ A are perpendicular, so $I_0 = \sqrt{0.075^2 + 0.4364^2} =$ **0.4427 A**. $P_0 = V^2/R_C = 240^2/3200 =$ **18.0 W**.
>
> c. Short circuit at full voltage: $|Z| = |1.1 + j3.5| = 3.669\ \Omega$, so $I = 240/3.669 =$ **65.42 A** and $P = I^2 R = 65.42^2 \times 1.1 =$ **4707 W** (a destructive overload, nearly 16

× rated current).

d. Rated input current $= 1000/240 =$ **4.167 A**. The supply voltage that drives it into the short circuit is $4.167 \times 3.669 =$ **15.29 V** (6.4 % of rated voltage).

These match the official answers.

PQ 3.28 (practice questions).

OC (from (b)): 240 V, 0.4427 A, 18.0 W. $R_C = V^2/P = 240^2/18.0 = 3200\ \Omega$. $S = 240 \times 0.4427 = 106.25$ VA, $Q = \sqrt{106.25^2 - 18^2} = 104.7$ VAr, so $X_M = V^2/Q = 550.1\ \Omega$.

SC (from (c)): 240 V, 65.42 A, 4707 W. $R = P/I^2 = 4707/65.42^2 = 1.100\ \Omega$. $|Z| = 240/65.42 = 3.669\ \Omega$, so $X = \sqrt{3.669^2 - 1.100^2} = 3.500\ \Omega$.

These recover the 3.27 values. The official answers are $R_C = 3300\ \Omega$, $X_M = 549.6\ \Omega$, $R = 1.100\ \Omega$, $X = 3.501\ \Omega$. The small differences in X_M and X come from rounding the 3.27 results. The 3300 Ω can't come from 18.0 W at 240 V ($240^2/18 = 3200$ exactly), so it is almost certainly a typo for 3200 Ω.

PQ 3.29 (practice questions, do (a)–(c)).

Hint: $V_1 = V_2' + I_2'(R + jX)$.

a. SC: $R = 55/9.5^2 =$ **0.6094 Ω**, $|Z| = 27/9.5 = 2.842\ \Omega$, $X = \sqrt{2.842^2 - 0.6094^2} =$ **2.776 Ω**. OC: $R_C = 240^2/45 =$ **1280 Ω**; $S = 288$ VA, $Q = \sqrt{288^2 - 45^2} = 284.5$ VAr, $X_M = 240^2/284.5 =$ **202.5 Ω**.

b. Rated output voltage referred to the primary: $V_2' =$ **240∠0° V**. $I_2' = 2400/240 = 10$ A at $\phi = -\cos^{-1} 0.87 = -29.54°$, so $I_2' =$ **10∠−29.54° A**.

c. $V_1 = 240 + 10\angle{-29.54°} \times (0.6094 + j2.776) = 259.9\angle 4.67°$ V, so $|V_1| =$ **259.9 V**.

d. is efficiency (pack M5). The official answers are $P_O = 2088$ W, $P_{CU} = 60.94$ W, $P_{FE} = 52.77$ W and $\eta = 94.84\%$.

These match the official answers.

2024 test Q6 (4 marks).

Hint: both tests are line values on a star connection, so convert to per-phase values first.

Per phase (SC): $V_1 = 710/\sqrt{3} = 409.9$ V, $I_1 = 390$ A, $P_1 = 140/3 = 46.67$ kW.

$R = P_1/I_1^2 = 46\,667/390^2 =$ **0.3068 Ω**. $|Z| = 409.9/390 = 1.051\ \Omega$, so $X = \sqrt{1.051^2 - 0.3068^2} =$ **1.005 Ω**.

I disagree with the answer sheet's $R = 0.3086\,\Omega$. $46\,667/390^2 = 0.3068$, so the sheet has transposed two digits (your Test Reference flags the same thing). Its $X = 1.005\ \Omega$ and its intermediate values ($V_1 = 409.9$ V, $P_1 = 46.67$ kW, $Q_1 = 152.9$ kVAr) agree with this working.

2024 test Q7 (4 marks).

Hint: generating Q means leading, so the angle is positive.

Referred to the primary, rated output voltage is the primary rating: $V_2' = 14\,000/\sqrt{3} =$ **8.083∠0° kV** (per phase, star assumed).

$|I_2'| = S/(3V_2') = 150 \times 10^6/(3 \times 8083) = 6186$ A. The load generates Q, so $Q = -35$ MVAr and $\sin(-\phi) = -35/150$, giving $\phi = +13.49°$. So $I_2' =$ **6.186∠+13.49° kA**.

These match the official answers.

2024 test Q8 (4 marks).

- **Ideal transformer.** The output voltage stays constant, because it is set only by the input voltage and the turns ratio and is not affected by the load current. The output current halves, by Ohm's law, because the load resistance has doubled.
- **Real transformer.** It has series winding resistance and leakage reactance. When the load current falls, the voltage drop across these series terms falls, so the output voltage rises slightly. The output current therefore falls to slightly more than half of its original value.

(Exam instruction: words only, so no equations or sketches.) These are the course's model points.

2023 quiz Q6 (9 marks).

a. $V_2' = 3300/\sqrt{3} =$ **1.905∠0° kV** per phase. $S = P/\text{pf} = 27/0.78 = 34.62$ kVA, so $|I_2'| = 34\,620/(3 \times 1905) = 6.056$ A at $-\cos^{-1} 0.78 = -38.74°$: $I_2' =$ **6.056∠−38.74° A**.

b. Circuit: R_C (18 kΩ) and jX_M (j2.3 kΩ) across V_1, then $4 + j35\ \Omega$ in series to V_2'. The shunt branch doesn't affect the KVL:

$$V_1 = V_2' + I_2'(R + jX) = 2100 + 7.7\angle -25° \times (4 + j35) = 2.254\angle 5.889°\ \text{kV per phase}$$

Line voltage: $\sqrt{3} \times 2.254 =$ **3.904 kV**.

I disagree with the sheet's 3.883 kV. $\sqrt{3} \times 2.254 = 3.904$, and the sheet's own phase value ($2.254 \angle 5.889^\circ$ kV) agrees with mine.

PQ 3.30 (practice questions, overflow: do (a), (b)).

a. $V_2' = 415/\sqrt{3} =$ **239.6∠0° V**. $I_2' = 3000/(3 \times 239.6) =$ **4.174∠0° A** (unity pf).

b. $V_1 = 239.6 + 4.174 \times (0.95 + j8.5) = 246.1 \angle 8.29^\circ$ V per phase, so the line voltage is $\sqrt{3} \times 246.1 =$ **426.3 V**. The official answer is 426.4 V ($V_1 = 246.2$ V); the difference is rounding.

(c)–(d) are pack M5. The official answers are $P_{CU} = 49.65$ W, $P_{FE} = 55.10$ W and $\eta = 96.62\%$.

PQ 3.31 (practice questions, overflow: do (a), (b)).

Hint: both tests are line values on a star connection, so per phase use $95/\sqrt{3}$ V and $12/3$ kW, then $415/\sqrt{3}$ V and $800/3$ W. Note that the OC test was **not** at rated voltage, so X_M is the low-flux value (see Example 3).

a. SC per phase: 54.85 V, 500 A, 4000 W. $R = 4000/500^2 =$ **0.01600 Ω**, $|Z| = 54.85/500 = 0.1097\ \Omega$, $X =$ **0.1085 Ω**. OC per phase: 239.6 V, 15 A, 266.7 W. $R_C = 239.6^2/266.7 =$ **215.3 Ω**; $S = 3594$ VA, $Q = 3584$ VAr, $X_M = 239.6^2/3584 =$ **16.02 Ω**.

b. $V_2' = 15\,000/\sqrt{3} =$ **8.660∠0° kV**. $|I_2'| = (50/0.8) \times 10^6/(3 \times 8660) = 2406$ A, so $I_2' =$ **2.406∠−36.87° kA**.
$V_1 = 8660 + 2406 \angle -36.87^\circ \times (0.016 + j0.1085) = 8.849$ kV per phase ($8.850 \angle 1.20^\circ$ kV unrounded), so the line voltage is **15.33 kV**.

c. is pack M5. The official answer is $\eta = 97.34\%$ ($P_{FE} = 1.091$ MW, $P_{CU} = 0.2779$ MW).

These match the official answers.

New questions

New Question 1 (easy) [6 marks].

a. $n = 8$. $Z_L' = 64(0.4 + j0.3) =$ $\mathbf{25.6 + j19.2\ \Omega}$ $(= 32 \angle 36.87^\circ\ \Omega)$.

b. $Z_{total} = 8 + 25.6 + j19.2 = 33.6 + j19.2 = 38.70 \angle 29.74^\circ\ \Omega$.

$I_1 = 240/38.70 \angle 29.74^\circ =$ $\mathbf{6.202 \angle -29.74^\circ\ A}$.

$V_1 = I_1 Z_L' = 6.202 \angle -29.74^\circ \times 32 \angle 36.87^\circ =$ $\mathbf{198.5 \angle 7.13^\circ\ V}$.

c. $I_2 = nI_1 = \mathbf{49.61\angle -29.74^\circ}$ **A**. $V_2 = V_1/n = \mathbf{24.81\angle 7.13^\circ}$ **V**.

$P = |I_2|^2 \times 0.4 = 49.61^2 \times 0.4 = \mathbf{984.6}$ **W**.

Check from the secondary side: refer the source resistance across, $8/64 = 0.125\,\Omega$, and use the source $240/8 = 30$ V. Then $I_2 = 30/(0.525 + j0.3) = 49.61\angle -29.74^\circ$ A. ✓

New Question 2 (medium) [5 marks].

$n = 415/110 = 3.773$, so $n^2 = 14.23$.

SC (already on the primary side): $S = 22 \times 12.05 = 265.1$ VA.
$Q = \sqrt{265.1^2 - 120^2} = 236.4$ VAr.

$R = 120/12.05^2 = \mathbf{0.8264\,\Omega}$. $X = 236.4/12.05^2 = \mathbf{1.628\,\Omega}$.

(12.05 A is the rated primary current, $5000/415$.)

OC (on the LV side): $S = 110 \times 1.8 = 198$ VA. $Q = \sqrt{198^2 - 55^2} = 190.2$ VAr.

$R_{C,LV} = 110^2/55 = 220\,\Omega$. $X_{M,LV} = 110^2/190.2 = 63.61\,\Omega$.

Refer to the primary: $R_C = 14.23 \times 220 = \mathbf{3131\,\Omega}$. $X_M = 14.23 \times 63.61 = \mathbf{905.5\,\Omega}$.

Answer circuit for New Question 2: 3131 Ω ‖ j905.5 Ω across the 415 V input, then 0.8264 Ω + j1.628 Ω

`CIRCUIT:` $3131\,\Omega \parallel j905.5\,\Omega$ across the 415 V input terminals, then $0.8264\,\Omega$ and $j1.628\,\Omega$ in series to V_2'.

(Check: 415 V across these shunt values draws 0.477 A on the HV side. That equals $1.8/n$. ✓)

New Question 3 (medium) [5 marks].

a. **OC, per phase:** $V_1 = 11000/\sqrt{3} = 6351$ V, $I_1 = 1.2$ A, $P_1 = 1900/3 = 633.3$ W.

$S_1 = 7621$ VA, $Q_1 = \sqrt{7621^2 - 633.3^2} = 7595$ VAr.

$R_C = 6351^2/633.3 = \mathbf{63.68\,k\Omega}$. $X_M = 6351^2/7595 = \mathbf{5.311\,k\Omega}$.

SC, per phase: $V_1 = 480/\sqrt{3} = 277.1$ V, $I_1 = 26.24$ A, $P_1 = 5200/3 = 1733$ W.

$S_1 = 7272$ VA, $Q_1 = 7062$ VAr.

$R = 1733/26.24^2 = \mathbf{2.517\,\Omega}$. $X = 7062/26.24^2 = \mathbf{10.26\,\Omega}$.

b. $I_{rated} = 500\text{k}/(\sqrt{3} \times 11\text{k}) = 26.24$ A. ✓ The SC voltage is $480/11000 = 4.4\%$ of rated, consistent with "about 10% or less".

New Question 4 (hard, chained) [8 marks].

a. $R'_L = n^2 R_L = 14.23 \times 2.42 = \mathbf{34.45\ \Omega}$.

b. $V'_2 = V_1 - I'_2(R + jX)$, with $R = R_1 + R'_2$ and $X = X_1 + X'_2$.

c. $$V'_2 = 415 \times \frac{34.45}{34.45 + 0.8264 + j1.628} = 415 \times \frac{34.45}{35.27 + j1.628}$$

$V'_2 = \mathbf{404.9 \angle -2.64^\circ\ V}$ (lagging V_1).

Actual: $V_2 = 404.9/3.773 = \mathbf{107.3\ V}$ (down from 110 V on no load).

d. $I'_2 = V'_2/R'_L = 11.75 \angle -2.64^\circ$ A.

$I_{sh} = 415/3131 + 415/(j905.5) = 0.1325 - j0.4583 = 0.4771 \angle -73.87^\circ$ A.

$I_1 = I'_2 + I_{sh} = \mathbf{11.92 \angle -4.82^\circ\ A}$.

e. $Z'_L = 34.45 \angle -36.87^\circ\ \Omega$. The divider gives $V'_2 = 418.3 \angle -3.02^\circ$ V. **The magnitude rises above 415 V, but V'_2 still lags.** The student is wrong. The phase of $V'_2/V_1 = Z'_L/(Z'_L + R + jX)$ is a lag whenever the load angle is less inductive than $\tan^{-1}(X/R) = 63^\circ$, which covers every resistive or capacitive load.

New Question 5 (hard, saturation) [5 marks].

a. **120 V:** $S = 6.0$ VA, $Q = \sqrt{6^2 - 2.2^2} = 5.582$ VAr. $R_C = 120^2/2.2 = \mathbf{6545\ \Omega}$. $X_M = 120^2/5.582 = \mathbf{2580\ \Omega}$.

240 V: $S = 100.8$ VA, $Q = 100.4$ VAr. $R_C = 240^2/8.6 = \mathbf{6698\ \Omega}$. $X_M = 240^2/100.4 = \mathbf{573.5\ \Omega}$.

b. Predicted: $I_M = 240/2580 = 0.093$ A and $I_C = 240/6545 = 0.037$ A, so $|I| = 0.100$ A. $P = 240^2/6545 = 8.8$ W.

Measured: 0.42 A and 8.6 W. **The loss prediction is good (within 2%), but the current is about 4 times too low.**

c. At 240 V the peak flux density is twice as high (back-emf equation), and the core is in saturation. The permeability falls, so L_M falls from 8.2 H to 1.8 H and the magnetising current rises sharply. Core loss still scales as V^2, so R_C stays roughly constant. Use the **rated-voltage (240 V) values**. This is the same effect as your Prac 2.

6. Quick recall check

Question

1. The applied voltage is small (about 10% of rated). The shunt currents are proportional to V, so they are negligible next to the short-circuit current, and the circuit reduces to $R + jX$ in series.
2. $n^2 Z_L = 100 \times 2 = 200\,\Omega$.
3. $X_M = V^2/Q$ (parallel). $X = Q/I^2$ (series). Both use $Q = \sqrt{(VI)^2 - P^2}$.
4. It lags. The series X makes I_2' lag V_1, and $V_2' = I_2' R_L'$ is in phase with I_2'.
5. $V_1 = 6600/\sqrt{3} = 3811$ V, $I_1 = 3$ A, $P_1 = 2$ kW.